# Venkata Lakshmi Parimala Pasupuleti

AI Platform Engineer · Generative AI · RAG · Agentic Frameworks · LLM Evaluation · Multi-Provider Orchestration · Python · Healthcare AI · HIPAA

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## PROFESSIONAL SUMMARY

AI platform engineer with 8 years building production enterprise systems and hands-on experience shipping **Generative AI, RAG, agentic, and LLM evaluation** solutions into regulated, data-sensitive environments. Built **Leasel** — a production agentic AI platform using **LangGraph, RAG architectures, prompt engineering, and LLM evaluation**: hybrid retrieval (TF-IDF + semantic reranking), citation grounding, conditional HITL routing, and a CI regression harness tracking accuracy, groundedness, and hallucination rate across 1,240 test cases. Built **AgentProbe** — an open-source **LLM evaluation and observability toolkit** with multi-provider orchestration across **Claude (Anthropic), OpenAI, Groq, and Ollama**; 5-metric eval engine; LLM-as-judge scoring; offline + online evaluation loops. Built **AI on FHIR** — an NLP-driven healthcare data platform integrating LLMs with FHIR APIs under **HIPAA-compliant RBAC**. At Entrata (8yr): shipped 20+ production modules, owned **HIPAA-compliant pipelines** (100% PHI compliance, zero security incidents), and mentored 5+ engineers across a complex multi-tenant healthcare-adjacent SaaS platform. Python, Java, TypeScript, React; REST APIs; Docker, Kubernetes; AWS; CI/CD (GitHub Actions).

## CORE COMPETENCIES

Generative AI / LLMs | Agentic Frameworks (LangGraph / LangChain) | RAG Architectures | Prompt Engineering & Structured Outputs | LLM Evaluation & Observability | LLM-as-Judge Scoring | Multi-Provider AI Orchestration | Guardrails & Hallucination Detection | Vector Databases & Semantic Search | Model Fine-Tuning (HuggingFace / PyTorch) | HITL / Human-in-the-Loop | Healthcare AI & FHIR Integration | HIPAA / PHI Compliance | Python / FastAPI | Java / Spring Boot | TypeScript / Node.js | React / Angular | REST APIs & Microservices | AWS (S3 / Lambda / Bedrock / SageMaker) | Docker / Kubernetes | CI/CD (GitHub Actions) | Multi-Tenant Architecture | PostgreSQL / SQLite / Redis | Regulated / Data-Sensitive Environments

## EXPERIENCE

### Blue Horse Digital

Sep 2026 - Present

**AI/ML Engineer (Volunteer) — Virginia, USA**

- Building AI-enabled applications and data pipelines; integrating LLM and agentic components with backend systems; surfacing health and business insights into application-facing APIs.

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- Built an **agentic workflow system** using LangGraph (parse → plan → retrieve → validate → route) with **RAG over a regulatory corpus**, cited-source grounding, validation guardrails, and MCP server deployment for downstream tool integration.

### Entrata — PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer — Lehi, UT (Remote)**

- **HIPAA-compliant data pipelines with 100% PHI compliance** — Libsodium-encrypted pipelines, multi-tenant data isolation at the schema layer, access controls per record type; zero security incidents across the full AWS S3 migration serving 100K+ monthly active users.
- **Shipped 20+ production modules end-to-end** (Python, Java, PHP, PostgreSQL, RabbitMQ) across 120+ Agile sprints; led microservices redesign (Docker) — 50% scalability improvement, 30% deployment time reduction.
- **Mentored 5+ engineers**; conducted 300+ code reviews; built internal debugging tools reducing resolution time by 40%; optimized 200+ SQL queries (30% performance gain).
- 80% reduction in manual operational workload via Python/Bash automation pipelines; owned async processing system (RabbitMQ) with retry logic and idempotency guarantees through production incidents.

### Entrata India — Db Xento Systems

Jan 2017 - Sep 2018

**Associate / Trainee Software Engineer — Pune, India**

- Built compliance and document management modules (PHP, Python, Java, PostgreSQL); led AWS S3 cloud migration; income-limits import (1.6M HUD/Tax Credit records across 5 years).

## PROJECTS

### Leasel: Production Agentic AI Platform — RAG, Evaluation & HITL

LangGraph · RAG · LLM Eval · Guardrails · HIPAA-pattern

Production GenAI platform applying **RAG architectures, prompt engineering, agentic frameworks, and LLM evaluation** to compliance workflows: hybrid retrieval (TF-IDF + semantic reranking), **citation grounding** to exact source clauses, **hallucination detection** via unsupported-claim flagging, missing/conflicting clause detection. **8-agent LangGraph StateGraph** with durable state, tool calling, and **conditional HITL routing** (fires when AI confidence is insufficient). Core design rule: LLM extracts and reasons; human or deterministic logic makes every regulated decision. **LLM evaluation pipeline**: 5-metric engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, offline + online eval loops, CI regression gate — 93.6% precision, 87.9% F1 on 1,240 test cases.

Python · LangGraph · FastAPI · TF-IDF · ChromaDB · OCR · SQLite · GitHub Actions · 2026

### AgentProbe: LLM Evaluation & Observability Toolkit

Multi-Provider · Eval Pipelines · Guardrails · CI/CD

Open-source toolkit for **LLM evaluation, observability, and regression testing**: **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), **LLM-as-judge scoring**, CI-integrated regression harness with golden datasets, offline + online evaluation loops, and **multi-provider AI orchestration** across Claude (Anthropic), OpenAI, Azure OpenAI, Groq, and Ollama. Prompt version management; reusable evaluation frameworks and tooling deployable in CI/CD. Implements guardrails and hallucination detection patterns applicable to healthcare AI contexts. Live: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · OpenAI · Anthropic · CI/CD (GitHub Actions) · 2025

### AI on FHIR: NLP-Driven Healthcare Data Integration

Healthcare AI · FHIR · HIPAA · LLMs · NLP

Healthcare AI platform integrating **LLMs, BERT, and embeddings** with enterprise FHIR APIs: translated natural-language queries (EN/ES) into structured FHIR API calls; enforced **HIPAA-compliant RBAC** across structured medical records; **~60% reduction in manual clinical retrieval time**. Demonstrates LLM integration into regulated healthcare data infrastructure — the same accountability standard Health100 requires.

Python · Flask · React · LLMs · BERT · FHIR APIs · Docker · spaCy · 2024

## EDUCATION

**M.S. Data Analytics Engineering — George Mason University, Virginia** May 2026 | GPA 3.87

**B.Tech Computer Science & Engineering — JNTU, Anantapur** 2017